

Name: Calvin, Andrew, Dylan **Date:** 2/15/2024

Class: Engineering Period 3

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The lens cover must protect the camera lens from scratches, dents, and fingerprints when attached to the camera.

This is the primary purpose of a camera lens, every single camera lens ever has a cover that protects the lens.

This is validated by Mr.O'Farrell, as this is pretty much the sole function of the product. Mr.O'Farrell, or whoever is using the cover lens, needs this to keep the camera safe from scratches or other things that would get trapped on it.

Design Requirement #2 (HIGH Priority)

The lens cover must stay securely on the camera lens without falling off and measure 3.52" long, and 3.01" in height.

These are the measurements that we took from Mr. O'Farrell's camera from the TV department.

This is validated by Mr.O'Farrell. The camera lens cover would not be able to complete its job if it would be constantly falling off. This is a very important design requirement.

Design Requirement #3 (HIGH Priority)

The lens cover must stay attached to the camera while not covering the lens so that it does not get lost.

We have seen many examples of camera lens covers being lost when not attached, as we are making this product to replace camera lens covers that have been lost. We plan to make ours harder to lose, as they will be attached to the camera even when in the "off" position and aren't actively being used. This can be done by being attached by a chord, hinge, or other ways.

The Stakeholder, Mr. O'Farrell and the rest of the people using the cameras would agree that this is an important design requirement. The reason that we are making these lens

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covers in the first place is because the TV department had lost its previous ones. Adding this design requirement could make this less likely to happen.

Design Requirement #4 (HIGH Priority)

The lens cover should be easy to put on and take off of the lens when the user wants to.

If the lens cover is hard to disconnect then the user might miss a shot they needed for whatever they are filming. If it is hard to attach then the user might not bother with it and leave the lens cover off, opening the lens to damage.

This would benefit Mr.O'Farrell because they would need to remove the lens quickly if they wanted to record something, in addition to likely not using the lens cover at all if it was too difficult to put back on.

Design Requirement #5 (HIGH Priority)

The cover should be durable, because only if it stays together does the lens stay safe.

The lens cover needs to be stronger than the lens because if it were to break after only a little bit of wear and tear it would be useless.

This is validated by Mr.O'Farrell. The camera lens cover would not be able to protect the camera if it was incapable of protecting itself.

Design Requirement #6 (LOW Priority)

The camera lens cover should be black in color, so that it fits in with the rest of the camera.

Mr.O'Farrell's cameras are black, and would find it more fitting if the colors match.

This is validated by Mr.O'Farrell. It is not a necessary design requirement, but it would look nicer and Mr.O'Farrell would like it more if we did.

Design Requirement #7 (LOW Priority)

The cover should weigh less than 3 pounds to not weigh down the camera too much.

This design requirement is just to make it easier to carry around and not be awkward when using it. Having a lighter cover would create a product that is easier to handle.

This is very low on the priority list because with the materials we are using (product is likely going to be 3d printed) and the size means that the only way the product could actually feel

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pretty hefty is if we made it huge, and because of the size of the cameras that the stakeholders are using, this is not going to happen.